

# RAG Evaluation Suite — Retrieval & Generation Metrics — Project Explanation

## Overview

This project builds a working RAG pipeline using LangChain, OpenAI embeddings, and a persistent Chroma vector store, then evaluates it with a full seven-metric DeepEval suite covering both retrieval quality and generation quality, using an LLM as an automated judge scored against ground-truth answers.

## What the Notebook Does

The pipeline loads a CSV knowledge base into Chroma with cosine similarity, retrieves using a similarity-score-threshold retriever that rejects weak matches rather than always returning a fixed count, and builds a RAG chain that returns both the generated answer and its source documents together. Three retrieval-side metrics, Contextual Precision, Recall, and Relevancy, are each tested against deliberately corrupted context to confirm they detect irrelevant, missing, or diluted retrieval rather than always scoring high.

The notebook's newest section adds four generation-side metrics. Answer Relevancy checks whether the response addresses the question independent of context. Faithfulness checks whether the answer stays grounded in what was actually retrieved. The Hallucination metric is tested twice, once against a reasonable model response and once against a deliberately fabricated answer containing nonsensical technical language, confirming the metric correctly flags obvious fabrication against the same ground-truth context. Finally, a custom GEval metric named RAG Fact Checker defines a five-step evaluation rubric in plain English, decomposing the answer into statements and checking each for relevance, correctness, grounding in retrieved context, and fabrication, demonstrating that domain-specific evaluation criteria can be expressed as a scored metric without building one from scratch.

## Skills Demonstrated

- Full-pipeline RAG evaluation spanning both retrieval and generation quality
- LLM-as-judge methodology using DeepEval's built-in and custom GEval metrics
- Distinguishing faithfulness from hallucination as related but separate failure modes
- Rigorous metric validation through deliberately corrupted or fabricated test inputs

This seven-metric evaluation suite is the direct blueprint for benchmarking both retrieval quality and answer trustworthiness in my dissertation's hybrid GraphRAG and HippoRAG agentic intelligence platform.